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Personal-Data Discovery and Privacy Evidence Inventory

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Problem Description

Personal data is scattered across spreadsheets and databases, making it difficult to maintain an accurate inventory. New fields may lack documented purposes or retention periods, while manual scans become outdated. Detection can also be inconsistent, and findings may lack measurable evidence.

- **Scattered data:** No centralized inventory across sources.
- **Missing metadata:** Purpose and retention details may be undocumented.
- **Detection gaps: New** fields can be missed by manual scans.
- **Limited evidence:** Findings need traceable support and measurable accuracy.

Proposed approach: Scan synthetic CSV files and one SQL database, flag documentation gaps, and maintain an evidence inventory for human review.

Background

Organizations need to know what personal data they hold, where it is stored, why it is collected, and how long it is retained. Spreadsheets and database schemas can change over time, making manually maintained inventories unreliable. DataPulse combines configurable data detection with purpose and retention metadata, scan history, and human review. It is designed as a focused prototype using synthetic data and a structured SQL source, with privacy documentation principles informing the inventory design [1]

Literature & RFC Review

- **GDPR, Articles 5 and 30 [1]:** Relevant principles include purpose limitation, storage limitation, and records of processing activities.
- **NIST SP 800-122 [2]:** Provides guidance for protecting personally identifiable information (PII).
- **Precision and recall [3]:** Provide measures for evaluating detection quality against manually labelled test data.

DataPulse applies these references to technical discovery and evidence tracking. It supports documentation and review; it does not determine legal compliance.

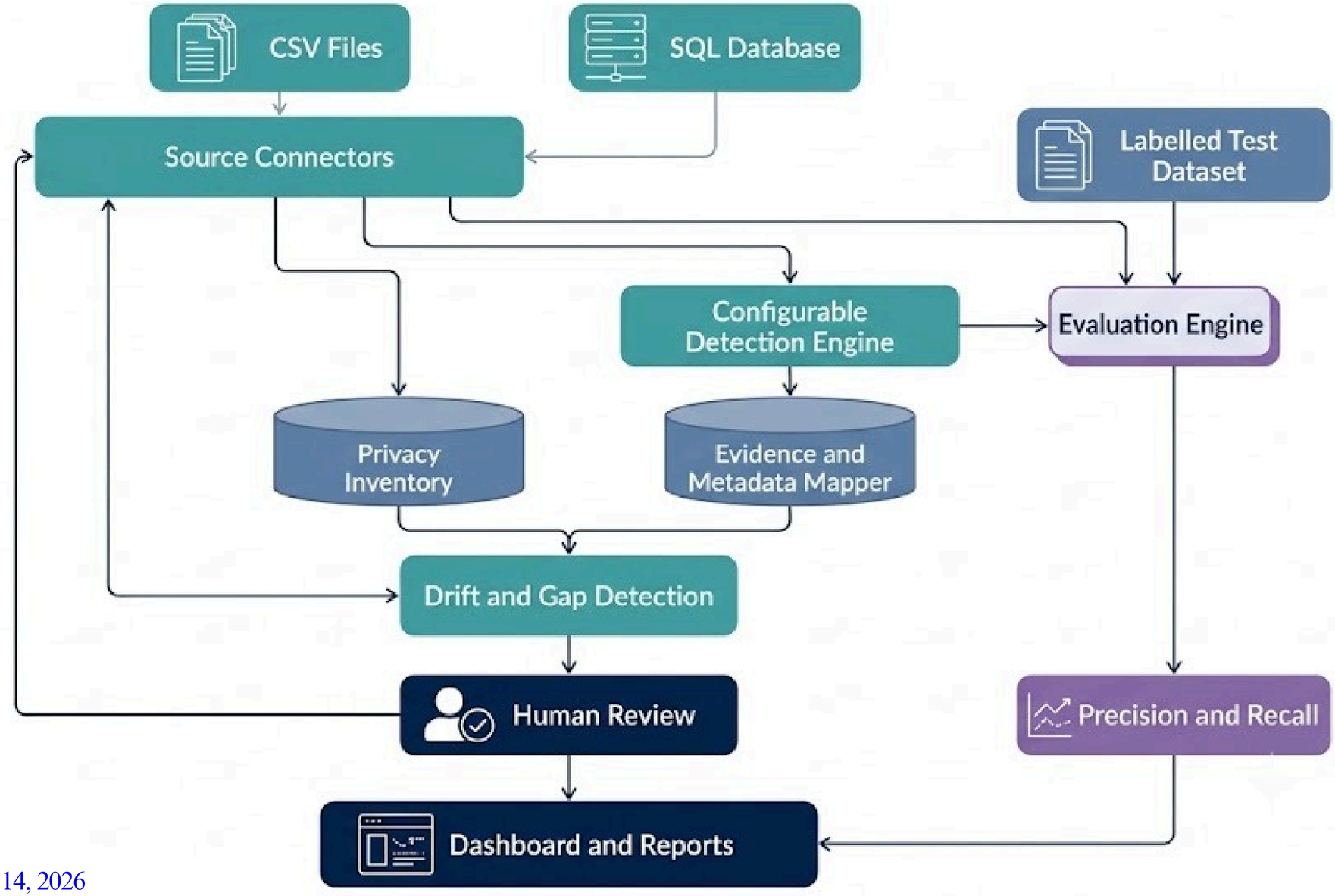
Proposed Solution

DataPluse scans synthetic CSV files and one SQL database using configurable column-name patterns, regular expressions, and value validation. It classifies candidate fields into five predefined data classes and stores their source locations, technical evidence, declared purposes, and retention metadata. Missing documentation is flagged for human review. Scan snapshots are compared to identify changes, while manually labelled synthetic test data is used to calculate per-class precision and recall. The prototype uses Python, SQLite, SQLAlchemy, and Streamlit.

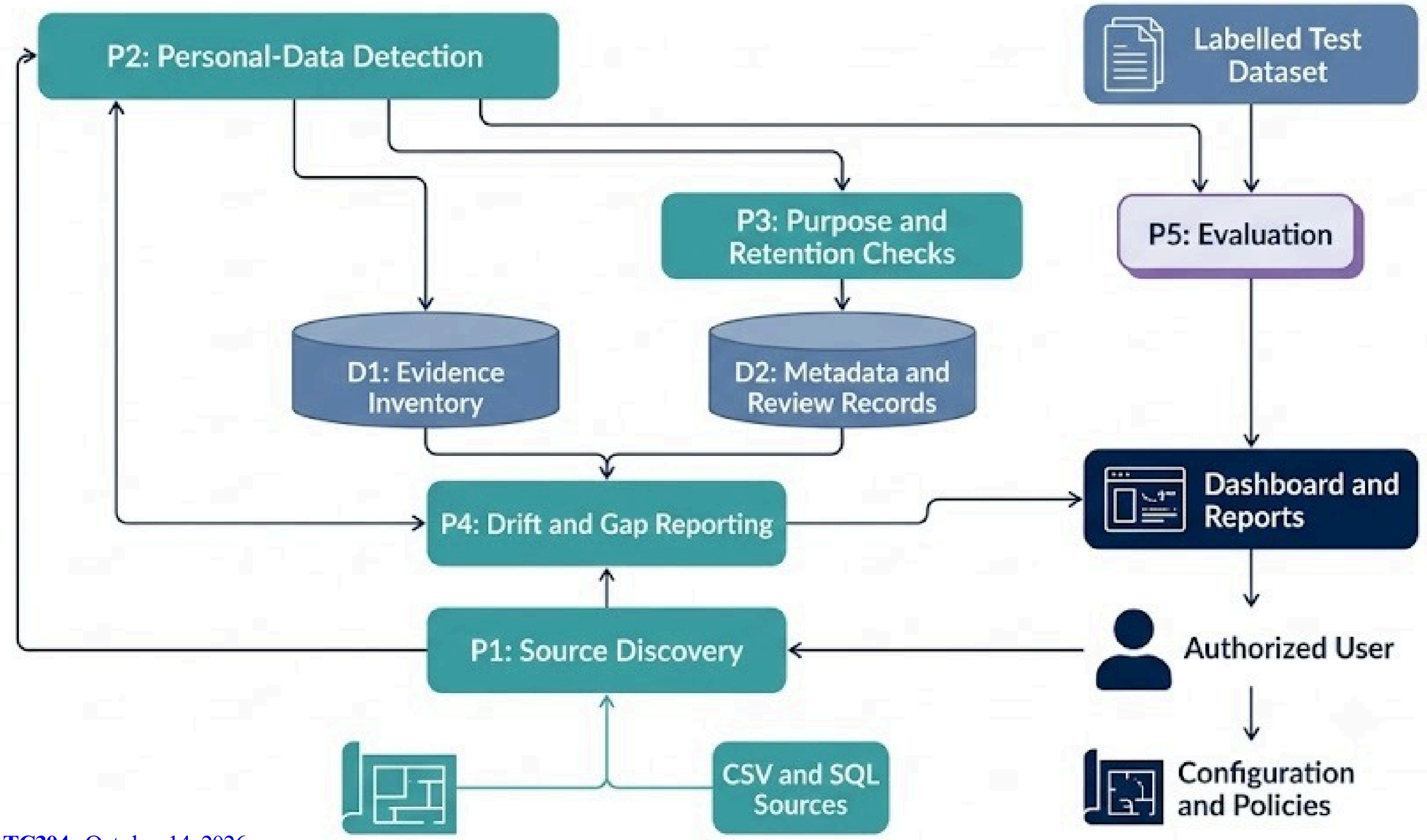
Optimization Proposed

Traditionally, personal-data discovery relies on separate source files and manual field identification, making it difficult to maintain an accurate inventory. DataPulse improves this process by creating a centralized inventory and applying configurable detection rules to identify candidate personal-data fields. Instead of leaving purpose and retention details undocumented, the system generates metadata-gap alerts for review. It also compares scan snapshots to identify changes and maintains review records, providing better evidence than one-time scans. Detection quality is evaluated using precision and recall on manually labelled test data. Human review and minimized exposure of raw data further improve the reliability and privacy of the proposed solution.

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

- **Phase 1: Foundation & Setup:** Set up the Python environment, create the project structure, prepare synthetic CSV datasets, and configure SQLite and SQLAlchemy for database connectivity. (17th to 19th October)
- **Phase 2: Source Integration & Detection:** Implement CSV and SQL connectors, configure detection rules using column-name patterns, regular expressions, and value validation, and define the five personal-data classes. (20th to 23rd October)
- **Phase 3: Inventory & Evaluation:** Develop the centralized inventory, purpose and retention metadata checks, snapshot comparison, and evidence-gap reporting. Evaluate detection performance using labelled synthetic data and measure precision and recall. (24th to 27th October)
- **Phase 4: Finalization & Demonstration:** Complete the Streamlit dashboard, test the integrated prototype, finalize visualizations and documentation, and prepare the final submission and demonstration. (28th to 31st October)

Bibliography I

1. European Union. General Data Protection Regulation (GDPR), Articles 5 and 30. https://eur-lex.europa.eu/eli/reg/2016/679/oj
2. NIST. Guide to Protecting the Confidentiality of Personally Identifiable Information (PII), SP 800-122. https://csrc.nist.gov/pubs/sp/800/122/final
3. Powers, D. M. W. “Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness & Correlation.” Journal of Machine Learning Technologies, 2(1), 2011, pp. 37–63.
4. Ministry of Electronics and Information Technology, Government of India. Digital Personal Data Protection Act, 2023. https://www.meity.gov.in/data-protection-framework
5. AIORI Hackathon 3. Problem Statement PS-019.